LAB 10

Strategy-Search Stage: Grid Search, Out-of-Sample Ranking and Decay

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	Strategy search
TSC reference	A2, A4, K2
Duration	20 min
Folder	labs/Lab10-strategy-grid-search/
Files	Lab10-strategy-grid-search/search_stage.py, pipeline/search.py, Lab10-strategy-grid-search/solution/search.py
Mode	Runs ONLINE with your Alpaca paper keys in labs/.env, or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

A junior analyst wants the pipeline to 'find the best parameters'. The head of research agrees on one condition: parameters are ranked on 2018-2021 only and judged on the untouched 2022-2025 period (including the 2022 bear market), and the stage must report how much of the in-sample result decays.

Objective

Build `python -m pipeline search` to grid-search three signal families, rank parameter sets in sample, measure out-of-sample decay, prefer a robust parameter region, and decide whether anything deserves promotion into config.yaml.

Before You Start

- Open a terminal in the labs folder and activate your virtual environment.
- Have your AI coding assistant open (for example Claude Code) in the labs folder if you want to rebuild the code from the prompts; the finished reference code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

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Step-by-Step Instructions

Step 1: Define the search space (A2, K2)

Read GRIDS in Lab10-strategy-grid-search/solution/search.py: sma_cross fast x slow (24 valid pairs), rsi_reversion n x entry x exit (36), bollinger_breakout window x k (16) - 76 sets.

Check: You can say why fast 50 / slow 50 is skipped and why every extra set increases the chance of a lucky winner.

Step 2: Build the grid search (A2, A4)

Ask your assistant for the search engine.

AI prompt - paste into your AI coding assistant
Create labs/pipeline/search.py. IS_END = '2021-12-31', OOS_START = '2022-01-01'. GRIDS = {'sma_cross': {'fast': [5,10,20,30,50], 'slow': [50,100,150,200,250]}, 'rsi_reversion': {'n': [2,3,5,14], 'entry': [10,20,30], 'exit': [50,60,70]}, 'bollinger_breakout': {'window': [10,20,30,50], 'k': [1.0,1.5,2.0,2.5]}}. grid_points(name) yields parameter dicts (skip sma_cross with fast >= slow). search(bars, costs) runs ONE backtest per set with pipeline.signals.target_position and common.backtest.backtest (signals are causal, so one run covers both windows), then records IS and OOS Sharpe and CAGR (common.metrics) and trades; add IS_rank and OOS_rank (method='min', NaN last) and decay = OOS - IS Sharpe. Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: search(...) returns a DataFrame with 76 rows.

Step 3: Add ranking, robustness and the decay report (A4, K2)

Ask your assistant for run() and main().

AI prompt - paste into your AI coding assistant
In labs/pipeline/search.py add robust_choice(ma) - for the sma_cross grid, the cell with the best AVERAGE in-sample Sharpe over itself and its neighbours - and run(symbol='SPY') for `python -m pipeline search`: print buy-and-hold Sharpe IS and OOS; a decay report with the best IS set per family (IS -> OOS Sharpe, decay, OOS rank of 76, trades); the top 5 by IS Sharpe; the Spearman rank correlation between IS and OOS Sharpe; the average decay of the top 5; the share of all sets that beat buy and hold OOS; the single best MA cell versus the robust MA region; and a suggested config.yaml strategy line (do NOT edit the config automatically). Save _outputs/Lab10_grid_results.csv, _outputs/Lab10_decay_report.csv and chart Lab10_grid_heatmap.png (IS and OOS Sharpe heatmaps of the MA grid with values in each cell). Rules: run from the labs folder; reuse labs/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline search --help shows --symbol.

Step 4: Run the search (A4)

Run: python -m pipeline search --symbol SPY (same as: python3 Lab10-strategy-grid-search/search_stage.py).

Check: 76 sets; buy-and-hold Sharpe 0.87 IS and 0.67 OOS. Best per family: sma_cross 10/100 IS 1.23 -> OOS 0.70 (decay -0.53, OOS rank 26); rsi_reversion n 2, entry 30, exit 70: 0.60 -> 0.74; bollinger_breakout 20 / 1.0: 0.91 -> 1.04 (OOS rank 3).

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Step 5: Read the overfitting evidence (A4, K2)

Read the top-5 table and the three summary lines.

Check: Top 5 are all sma_cross; average decay -0.62 Sharpe points; IS/OOS rank correlation 0.06 (the in-sample ranking has almost no predictive value); 43% of all sets beat buy and hold out of sample.

Step 6: Prefer a robust region to a single peak (A2, A4)

Compare the single best MA cell with the robust MA region and open _outputs/Lab10_grid_heatmap.png.

Check: Best cell 10/100: IS 1.23 -> OOS 0.70; robust region 5/50: neighbour-average IS 1.04, own IS 0.92 -> OOS 0.44 - robustness reduces luck but guarantees nothing.

Step 7: Decide whether to promote anything (A2, K2)

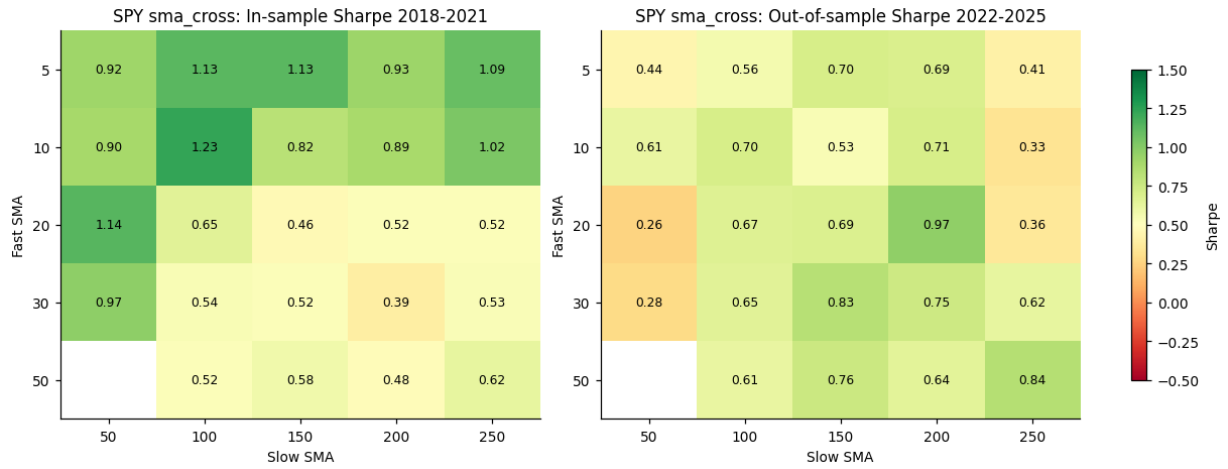
Do not edit config.yaml yet. Weigh the evidence first.

AI prompt - paste into your AI coding assistant
Using labs/_outputs/Lab10_decay_report.csv and labs/_outputs/Lab10_grid_results.csv, argue for and against replacing the pipeline's configured ema_cross 10/30 with the best sma_cross or bollinger_breakout set. Mention the number of sets tried (76), the decay, the rank correlation and what the diagnostics stage (Lab 18) should check before any change. Finish with a one-line recommendation.

Check: A balanced recommendation that keeps ema_cross until a diagnostics run and an A/B test (Lab 13) support a change.

Expected Output

python -m pipeline search --symbol SPY: 76 parameter sets; buy-and-hold Sharpe 0.87 in sample and 0.67 out of sample; best sma_cross (10/100) 1.23 -> 0.70; best rsi_reversion (2, 30, 70) 0.60 -> 0.74; best bollinger_breakout (20, 1.0) 0.91 -> 1.04; top-5 average decay -0.62; IS/OOS rank correlation 0.06; 43% of sets beat buy and hold out of sample; files _outputs/Lab10_grid_results.csv, _outputs/Lab10_decay_report.csv and chart _outputs/Lab10_grid_heatmap.png.



Lab 10 output (Lab10_grid_heatmap.png), offline mode on the bundled data.

Debrief

Grid search is a statistical search over model structures (A2, A4); comparing in-sample with out-of-sample results (K2) exposes how much of the 'best' result was luck. A hold-out period, a count of trials and a preference for robust regions are the pipeline's defences against fooling itself.

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Troubleshooting

Symptom	What to do
IntCastingNaNError on ranks	Some sets never trade in one window (NaN Sharpe); rank with na_option='bottom'.
Heatmap has a blank cell	Expected: fast 50 / slow 50 is not a valid pair.
Different results online	Alpaca's IEX data differ slightly from the bundled data; compare patterns, not decimals.

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